

Muhammad Faheem

+92 331-9090179 | faheemgurkani@gmail.com

muhammad-faheem | faheemgurkani | faheemgurkani.github.io

EXPERIENCE

National Centre for Physics (NCP) Research Intern

Islamabad, Pakistan
July 2024 – July 2025

- Conducted a comparative analysis of classical and deep learning-based object tracking methodologies, with primary focus on single-object tracking (SOT) architectures, frameworks, and evaluation protocols in research-driven environments
- Researched deep learning-based fault detection and anomaly detection techniques for HVAC systems, leading to an accepted IEEE conference paper titled *“Deep Learning-Based Fault Detection Framework for Predictive Maintenance in HVAC Systems”* at the 13th IEEE Conference on Technologies for Sustainability (SusTech 2026)

National Aerospace Science & Technology Park (NASTP) AI Engineering Intern

Rawalpindi, Pakistan
June 2025 – August 2025

- Designed and implemented an LLM-assisted OCR pipeline for business card digitization, enabling structured entity extraction (names, roles, organizations, contact details) and integration with an internal contact management system
- Conducted a research-focused implementation of an offline, gRPC-based multi-person behavior analysis system, integrating object detection, multi-object tracking (MOT), human pose estimation, and spatio-temporal action recognition for controlled evaluation and benchmarking

PROJECTS

BastionFed: Federated Learning & SOC Platform (Healthcare / IoMT)

- Built a full-stack SOC platform (**Next.js 14 + FastAPI/SSE**) with alert triage, incident forensics, FL model zoo, hash-chained audits, and an LLM copilot spanning **5+ analyst workflows**.
- Trained federated FV-DNN (image branch + meta-fusion) with multimodal drift detection, achieving **ROC-AUC ≥ 0.93** on global splits and authored a blue-team spec over **Firestore/GCP** (SQL, object storage, Redis).

Action Segmentation Using MS-TCN

- Engineered a multi-stage temporal convolutional network (MS-TCN) in PyTorch for frame-level action segmentation, achieving **+13.8% F1@50** over single-stage TCN baselines on long untrimmed video sequences
- Optimized training with mixed precision (AMP) and GPU-aware batching, enabling **sub-4.8 ms/frame inference** and stable convergence on sequences exceeding **10k frames/video**

Driver Assistance System Using RT-DETR

- Designed a real-time ADAS pipeline using fine-tuned RT-DETR, achieving **+18.6% mAP@50** over baseline DETR on dashcam-style data, with end-to-end inference latency of **3845 ms/frame** on RTX-class GPUs
- Deployed a production-grade inference stack using FastAPI (ASGI) and Next.js, supporting **1080p video streams at 25 FPS** with REST-based analytics endpoints

EDUCATION

National University of Computer and Emerging Sciences B.S. Artificial Intelligence

Islamabad, Pakistan
Aug 2022 – Present

- Honors: Deans List (Spring 2023, Fall 2025) | GPA: 3.21/4.00
- Relevant Coursework: Generative AI, Agentic AI, Reinforcement Learning, Computer Vision, ANN, Machine Learning, Digital Image Processing, NLP, Data Structures, Algorithms

SKILLS

Programming & Frameworks: Python, C/C++, JavaScript, TypeScript, FastAPI, React/Next.js

AI & ML: PyTorch, TensorFlow, Keras, scikit-learn, YOLO, ONNX, TensorRT, Hugging Face, LangChain, LangGraph, n8n

Data & Systems: NumPy, Pandas, Git, Docker, Linux, AWS, GCP, MCP